

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	11 September 2026
Team ID	NA
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Station Master Management	• Creation and configuration of custom station table (u_metro_station_details) • Management of operational station fields (u_station_name) • Ingestion of metro network station nodes (Ameerpet, Madhapur, LB Nagar, Kukatpally, Jubilee Hills)
FR-2	Service Catalog Booking Item	• Setup of catalog item Book A Metro Ticket under Service Catalog (Services category) • Definition of reference lookup fields for Starting From and Going To stations • Configuration of journey parameter variables (Type of Journey, No of Passengers)
FR-3	Dynamic Fare Calculation	• Real-time onChange Client Script on no_of_passengers • Automated route matrix mapping between origin and destination sys_id values • Dynamic fare split calculation for Single Journey (type_of_journey = 1) and Return Journey (type_of_journey = 2)
FR-4	Payment Mode Selection & Conditional UI	• Variable definition for payment mode choices (Cash, UPI, Credit/Debit Card, Others) • Configuration of Catalog UI Policy triggering on Mode of Payment = Others • Dynamic visibility and mandatory validation of field Enter payment mode
FR-5	Digital Ticket & QR Token Generation	• onSubmit Client Script intercepting checkout confirmation • Dynamic invocation of external QR code engine (api.qrserver.com) with unique ticket sys_id payload • Deployment of custom Service Portal widget (metro_qr_widget) to render a 250x250 scannable barcode via spModal
FR-6	Request Generation & Fulfillment Tracking	• Generation of ServiceNow parent Request (sc_request) and Requested Item (sc_req_item) records • Automated mapping of portal variables into backend ticket databases • Role-based access control rules (ACLs) restricting write access while allowing user station read lookups

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The booking interface on the Service Portal (/sp) must be fully responsive across mobile and desktop browsers, requiring no more than 4 form interactions to select a route, calculate fare, and display the QR pass.
NFR-2	Security	System must enforce table-level Access Control Lists (ACLs) on u_metro_station_details to permit read access to portal commuters while strictly restricting create, write, and delete permissions to authorized metro administrators (metro_admin).
NFR-3	Reliability	The QR code generation process must use fallback error handling in the Service Portal widget to ensure the modal window renders consistently across all user sessions without client-side DOM sandbox script errors.
NFR-4	Performance	The client-side dynamic fare computation script must execute and auto-populate amount fields in less than 200 ms upon changing station or passenger count variables.
NFR-5	Availability	The digital ticketing system deployed on the ServiceNow cloud instance must maintain high platform availability (99.9% uptime) to support uninterrupted commuter self-service during peak transit hours.
NFR-6	Scalability	The architecture must easily scale to accommodate future line extensions across hundreds of new station nodes and integrate downstream with SMS/WhatsApp notification APIs and payment gateway webhooks.